

5.BOM with part identifiers, ratings, datasheets, and cost/source

Item No.	Component Description	Exact Part	Verified Specifications	Qty	Cost (THB)	Source
1	Source Power Supply Unit	PLENTY COMPUTER ATX500WS	Input: 115V/230VAC ; Output: +3.3V(22A), +5V(14A), +12V1(13A), +12V2(14A), -12V(0.8A), +5VSB(2.5A); Max 500W	1	0.00	Course Provided
2	Adjustable Buck-Boost Power Module	XL6009	Step-up/Step-down DC-DC converter with CC/CV trim pots and heatsink; Input: 5 – 32V, Output: 1.25 – 35V/3A max	1	85.00	Workshop Stock
3	Dual Voltmeter & Ammeter Display Panel	DSN-VC288 V1.0	0.28" Dual LED display (0 – 100VDC/0), integrated current shunt, header cabling included	1	65.00	Workshop Stock

4	Glass Fuse Set	WIREMAN 5 AMP 20 MM	5 × 20mm Fast-acting glass tube fuses, 5A/250V breaking capacity	1box	35.00	Workshop Stock
5	Panel LED Indicator Holder & Lens	YZ659	Metal panel-mount bezel housing with locking ring for 5mm LED state indicators	2	30.00	Workshop Stock
6	Insulated Banana Jack Binding Posts	BP-4MM-I NS	4mm Female banana plug insulated binding posts (Red, Black, Yellow, Blue, Orange)	8	96.00	Workshop Stock
7	Acrylic Sheet	Clear Acrylic Sheet	3mm thick high-impact acrylic sheet for custom laser-cut/drilled bench enclosure box	1set	150.0 0	Workshop Stock

8	Power Dummy Load Resistors	Wirewound Power Resistors	High-power cement/metal-clad power resistors for rail loading/testing ($10\Omega/10W - 25$)	2	50.00	Workshop Stock
9	Maintained Control Switch	MTS-101 Toggle Switch	Single-pole single-throw toggle switch, rated $250VAC/3A$ for PS_ON control	1	10.00	Workshop Stock
10	Wiring, Terminal Ferrules & Heat Shrink	18 AWG Wire	18AWG stranded wire for main power, heat-shrink insulation tubing, and cable zip-ties	1set	40.00	Workshop Stock
total	-	-	-	-	561.0	-